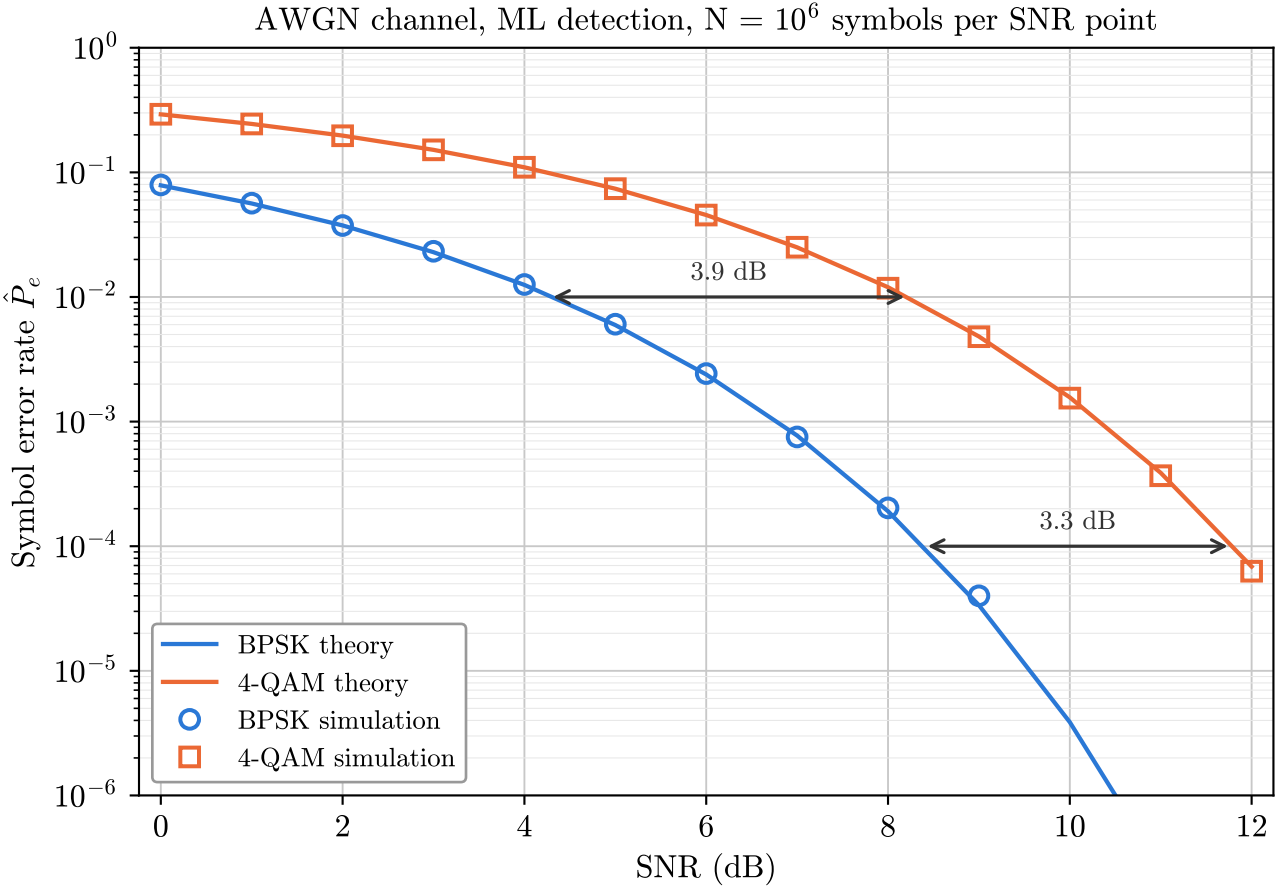
AWGN channel, ML detection, N = 10^6 symbols per SNR point
Symbol error rate $\hat{P}_e$
SNR (dB)
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
2
4
6
8
10
12
3.9 dB
3.3 dB
BPSK theory
4-QAM theory
BPSK simulation
4-QAM simulation